Smart Calculator

User Directions & Operation Guide

Getting Started

When powered on, the display briefly shows "Smart Calculator" then clears, ready for input. All operations follow the same pattern: enter a number, select an operation, enter a second number (if needed), then press # to calculate.

Basic Operations

Addition (+)

1. Type your first number using the digit keys (0-9)
2. Press A — the display shows +
3. Type your second number
4. Press # to see the result

Example: 5 A 3 # → Result: 8

Subtraction (-)

5. Type your first number
6. Press B — the display shows -
7. Type your second number
8. Press # to see the result

Example: 9 B 4 # → Result: 5

Multiplication (x)

9. Type your first number
10. Press C once — the display shows x
11. Type your second number
12. Press # to see the result

Example: 6 C 7 # → Result: 42

 Pressing C twice in a row activates Power Mode instead of multiplying.

Division (/)

13. Type your first number
14. Press D once — the display shows /

15. Type your second number
16. Press # to see the result

Example: 8 D 2 # → Result: 4

- *Dividing by zero will show a "Div by 0" error and reset the calculator.*
- *Pressing D twice in a row activates SQRT Mode instead of dividing.*

Factorial (!)

17. Type a number (must be 12 or less)
18. Press * — the display shows !
19. Press # to see the result

Example: 5 * # → Result: 120

- *Numbers greater than 12 will cause an Overflow error.*

Special Modes

The calculator has three special modes for advanced operations. Modes must be activated BEFORE entering numbers.

Power Mode (a^b)

20. Press C twice quickly — display briefly shows "Power Mode"
21. Type the base number
22. Press C once — the display shows ^
23. Type the exponent
24. Press # to see the result

Example: C C then 2 C 8 # → Result: 256

- *You must activate Power Mode before typing any numbers.*

Square Root Mode (√a)

25. Press D twice quickly — display briefly shows "SQRT Mode"
26. Type the number
27. Press D once — the display shows sqr
28. Press # to see the result

Example: D D then 1 6 D # → Result: 4

- *You must activate SQRT Mode before typing any numbers.*

Percent Mode (a x b / 100)

29. Type your first number
30. Press C, then press * — display briefly shows "Percent Mode" then restores the first number
31. Press D — the display shows %
32. Type the percentage value
33. Press # to see the result

Example: 2 0 C * then D 5 0 # → Result: 10 (50% of 20)

Resetting the Calculator

Press A twice in a row from anywhere to fully reset the calculator. The screen will briefly show "Reset" and all state, modes, and numbers will be cleared. This works at any point during a calculation.

Mode Quick Reference

Mode	Activation	Operator Key	Use Case
Normal	Default	A / B / C / D / *	+, -, x, /, !
Power	Press C C	C	a to the power of b
SQRT	Press D D	D	Square root of a
Percent	(num) then C then *	D	a x b / 100